

AERO70038 - Problem set 1

Hover and axial climb

1. Calculate the power required to hover at sea level (S.L.) for the single main rotor helicopter defined by the characteristics below, under the following assumptions:

- The main rotor profile power is 25% of the total main rotor power.
- Download on the airframe is 5% of the weight.
- Power absorbed by the tail rotor and transmission system is equal to 15% of the total main rotor power.

Helicopter characteristics:

Parameter	Value
Weight	44,500 N
Rotor radius	6.4 m
Air density at S.L.	1.2258 kg/m ³

$$P_h = 4P_o = \frac{20}{3} P_T$$

$$F_{\text{download}} = 5\% W = 2225 \text{ (N)}$$

$$P_h = T_h v_i = m w v_i = 2 \rho A v_i^3 = \frac{T_h^{\frac{3}{2}}}{\sqrt{2 \rho A}}$$

where

$$T_h = W + F_{\text{download}} = 46725 \text{ (N)}$$

$$A = \pi R^2 = 40.96 \pi \text{ (m}^2\text{)}$$

$$\therefore P_h = 568648 \text{ (W)} \quad \therefore P_{\text{main}} = \frac{P_h}{1-25\%} = 758197 \text{ (W)}$$

~~$$\therefore P_{\text{total}} = k P_h + P_o + P_T = (1.15 + 0.25 + 0.15) P_h$$~~

~~$$= 1.55 P_h = 881406 \text{ (W)}$$~~
$$P_{\text{total}} = (1 + 15\%) P_{\text{main}} = 871927 \text{ (W)}$$

2. Using Bernoulli's equation (instead of the energy equation) show that the induced velocity in the fully contracted wake of a rotor climbing with vertical velocity V_c is twice the induced velocity in the rotor plane.

From conservation of mass,

$$\dot{m} = \rho A (V_c + v_i) = \rho A_\infty (V_c + w)$$

From conservation of momentum,

$$T = \dot{m}(V_c + w) - \dot{m}V_c = \dot{m}w = \rho A (V_c + v_i) w$$

From Bernoulli's equation $\Delta p = \frac{T}{A} = p_2 - p_1 = (p_{\text{total}} - p_\infty) - (p_{\text{total}} - p_0)$

$$p_{\text{total}} = p_0 + \frac{1}{2}\rho V_c^2 = p_d + \frac{1}{2}\rho (V_c + v_i)^2 = p_\infty + \frac{1}{2}\rho (V_c + w)^2$$

$$\therefore T = (p_0 - p_\infty) A$$

$$\therefore T = \frac{1}{2}\rho A [(V_c + w)^2 - V_c^2] = \rho A (V_c w + \frac{1}{2}w^2)$$

Therefore,

$$(V_c + v_i) w = V_c w + \frac{1}{2}w^2$$

$$\therefore v_i w = \frac{1}{2}w^2$$

$$\therefore w = 2v_i$$

3. A tandem rotor helicopter with a gross weight of 8,845.2 kg has a rotor diameter of 13.72 m, a blade chord of 0.508 m, 3 blades and a tip speed of 213.43 m/s.

$$R = 6.86 \text{ m}$$

- (a) Estimate the total shaft power required to hover if the induced power factor for the front rotor is 1.20 and for the rear rotor 1.15. The rotor aerofoil has a zero lift drag coefficient of 0.01.
- (b) Estimate the installed power if transmission losses amount to 5% and the helicopter must demonstrate a rate of climb of 304.9 m/min (1,000 ft/min) at sea level.

$$(a) C_T = \frac{T}{\rho A V_{tip}^2} = \frac{mg}{2\rho\pi R^2 V_{tip}^2} = 5.255 \times 10^{-3}$$

$$C_{p, ideal} = C_T \lambda_i = \frac{C_T^{\frac{2}{3}}}{\sqrt{2}} = 2.694 \times 10^{-4}$$

$$C_{p0} = \frac{1}{8} 6 C_{d0} = \frac{1}{8} \left(\frac{N_b c}{\pi R} \right) C_{d0} = 8.839 \times 10^{-5}$$

$$\therefore C_{p, total} = (1.2 C_{p, ideal} + C_{p0}) + (1.15 C_{p, ideal} + C_{p0})$$

$$= 8.099 \times 10^{-4}$$

$$\therefore P_{total, h} = \rho (\pi R^2) V_{tip}^3 C_{p, total} = 1429493 \text{ (W)}$$

$$(b) \frac{P}{P_h} = \frac{V_c + v_i}{v_h} = \frac{V_c}{v_h} + \left(-\frac{V_c}{2v_h} + \sqrt{\left(\frac{V_c}{2v_h}\right)^2 + 1} \right) = \frac{V_c}{2v_h} + \sqrt{\left(\frac{V_c}{2v_h}\right)^2 + 1}$$

where

$$v_h = \sqrt{\frac{T}{2\rho A}} = \sqrt{\frac{mg}{4\rho(\pi R^2)}} = 10.94081 \text{ (m/s)}$$

$$V_c = 304.9 \text{ m/min} = 5.082 \text{ m/s}$$

$$\therefore P = 1.259 (P_h)_{total} \text{ where } P_h = T v_h = \frac{mg}{2} v_h = 474675 \text{ (W)}$$

$$\therefore P_{install} = 95\% P = 1.196 P_h = 1709537 \text{ (W)}$$

$$\therefore \Delta P = P - (P_h)_{total} = 0.259 (P_h)_{total} = 0.259 (2P_h)$$

$$\therefore P_{total} = (P_{total, h} + P) (1 + 5\%) = 1759143 \text{ (W)}$$

4. For a rotor with rectangular, untwisted blades, the inflow distribution is approximately triangular. Assuming no tip losses:
- Compute the variation with radius of inflow angle and section angle of attack for this rotor.
 - Derive the relation between the blade element lift coefficient and C_T/σ .
 - Derive an expression for the hovering power of this rotor in terms of C_T/σ and C_{d0} .

(a) $b = \text{constant}$

$\theta = \text{constant}$

according to the BET,

$$\begin{aligned} dC_T &= \frac{dT}{\rho A V_{tip}^2} = \frac{N_b dF_z}{\rho (\pi R^2) \Omega^2 R^2} = \frac{N_b dL}{\rho (\pi R^2) \Omega^2 R^2} = \frac{N_b (\frac{1}{2} \rho U^2 c C_l dy)}{\rho (\pi R^2) \Omega^2 R^2} \\ &= \frac{N_b (\Omega y)^2 c C_l dy}{2 \pi \Omega^2 R^3} = \frac{1}{2} b C_l r^2 dr \end{aligned}$$

according to BEMT, for $\lambda = \frac{V_c + V_i}{\Omega R}$, $\lambda_i = \frac{V_i}{\Omega R}$ and $\lambda_c = \frac{V_c}{\Omega R}$

$$\begin{aligned} dC_T &= \frac{m \dot{w}}{\rho A V_{tip}^2} = \frac{\rho dA (V_c + V_i) (2V_i)}{\rho \pi \Omega^2 R^3} = \frac{2 (2\pi y dy) (V_c + V_i) V_i}{\pi \Omega^2 R^3} \\ &= 4 \lambda \lambda_i r dr = 4 \lambda (\lambda - \lambda_c) r dr \end{aligned}$$

Therefore,

$$\frac{1}{2} b C_l r^2 = \frac{1}{2} b C_{l\alpha} (\theta - \phi) r^2 = \frac{1}{2} b C_{l\alpha} (\theta r^2 - \lambda r) = 4 (\lambda^2 - \lambda \lambda_c) r$$

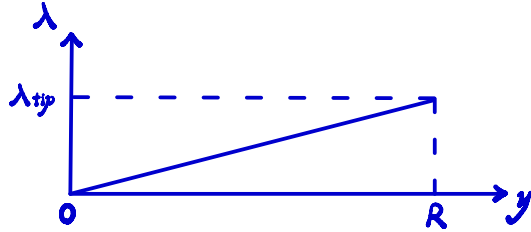
$$\therefore \lambda^2 - \lambda_c \lambda = \frac{1}{8} b C_{l\alpha} (\theta r - \lambda)$$

$$\therefore \lambda^2 + (\frac{1}{8} b C_{l\alpha} - \lambda_c) \lambda - \frac{1}{8} b C_{l\alpha} \theta r$$

$$\therefore \lambda = -(\frac{1}{16} b C_{l\alpha} - \frac{1}{2} \lambda_c) + \sqrt{(\frac{1}{16} b C_{l\alpha} - \frac{1}{2} \lambda_c)^2 + \frac{1}{8} b C_{l\alpha} \theta r}$$

$$\lambda = \frac{V_c + v_i}{\Omega R} = \frac{V_c + v_i}{\Omega y} \frac{y}{R} = \frac{U_p}{U_\tau} r = \phi r$$

for triangular inflow distribution, $\lambda = \lambda_{tip} r$



$$\therefore \phi = \frac{\lambda}{r} = \lambda_{tip} = \text{constant}$$

$$\therefore \theta = \theta_0 = \text{constant}$$

$$\therefore \alpha = \theta - \phi = \theta_0 - \lambda_{tip} = \text{constant}$$

$$(b) dC_T = \frac{1}{2} 6 C_{l\alpha} r^2 dr$$

$$= \frac{1}{2} 6 C_{l\alpha} (\theta r^2 - \lambda r) dr, \text{ assume symmetric airfoil } (\alpha_0 = 0)$$

$$= \frac{1}{2} 6 C_{l\alpha} (\theta_0 - \lambda_{tip}) r^2 dr$$

$$\therefore C_T = \frac{1}{2} 6 C_{l\alpha} (\theta_0 - \lambda_{tip}) \int r^2 dr = \frac{1}{6} 6 C_{l\alpha} (\theta_0 - \lambda_{tip}) = \frac{1}{6} 6 C_l$$

$$\therefore C_l = 6 \frac{C_T}{6}$$

$$(c) dC_P = \frac{1}{2} 6 (\phi C_l + C_{d0}) r^2 dr = \frac{1}{2} 6 C_l \lambda_{tip} r^2 dr + \frac{1}{2} 6 C_{d0} r^2 dr$$

$$\therefore C_P = \frac{1}{8} 6 C_l \lambda_{tip} + \frac{1}{8} 6 C_{d0}$$

$$= \frac{3}{4} 6 \left(\frac{C_T}{6} \right) \lambda_{tip} + \frac{1}{8} 6 C_{d0}$$

5. Calculate the collective pitch and power required for a helicopter with the characteristics below (neglect the effects of tail rotor power, transmission power losses, and airframe vertical drag; assume the blade has zero twist and neglect tip loss effects):

- (a) To hover at S.L.
 (b) To climb vertically at 3 m/s at S.L.

Helicopter characteristics:

Parameter	Value
Weight	27,460 N
Number of blades	2
Rotor rotational speed	372 RPM
Blade radius	5.4 m
Blade chord	0.33 m
Blade profile drag coefficient (C_{D0})	0.01
Blade section lift curve slope	5.73

$$\theta = \theta_0 = \text{constant}$$

$$\text{assuming uniform inflow, } \lambda = \lambda_i + \lambda_c = \text{constant}$$

$$\text{assuming symmetric airfoil, } \alpha_0 = 0$$

$$\begin{aligned} C_T &= \int_0^1 \frac{1}{2} \rho C_{l\alpha} (\theta_0 - \phi) r^3 dr \\ &= \frac{1}{2} \rho C_{l\alpha} \int_0^1 \theta_0 r^3 - \lambda r dr \\ &= \frac{1}{2} \rho C_{l\alpha} \left(\frac{1}{3} \theta_0 - \frac{1}{2} \lambda \right) \end{aligned}$$

$$\begin{aligned} C_P &= \int_0^1 \frac{1}{2} \rho (\phi C_l + C_d) r^3 dr \\ &= C_T \lambda + \frac{1}{8} \rho C_{d0} \end{aligned}$$

where

$$W = 27460 \text{ (N)}$$

$$\Omega = 372 \text{ (RPM)} = \frac{2\pi}{60} \cdot 372 \text{ (rad/s)} = 12.4\pi \text{ (rad/s)}$$

$$\phi = \frac{N_b C}{\pi R} = \frac{11}{90\pi}$$

$$\rho A \Omega^2 R^2 = \rho \pi \Omega^2 R^4 = 4965970 \text{ (N)}$$

$$\rho A \Omega^3 R^3 = \rho \pi \Omega^3 R^5 = 1044646694 \text{ (W)}$$

$$(a) C_T = \frac{T}{\rho A \Omega^2 R^2} = 2\lambda_h^2 = 5.52963 \times 10^{-3}$$

$$\therefore \lambda_h = \sqrt{\frac{C_T}{2}} = 0.05258$$

$$\therefore v_h = 11.06 \text{ (m/s)}$$

$$\therefore \theta_0 = 0.2277 \text{ (rad)} = 13.05^\circ$$

$$\therefore C_p = 3.3938 \times 10^{-4}$$

$$\therefore P = C_p \cdot \rho A \Omega^3 R^3 = 354531 \text{ (W)}$$

$$(b) V_c = 3 \text{ (m/s)}$$

$$\therefore \lambda_c = \frac{V_c}{\Omega R} = 0.0142612$$

$$\therefore \frac{v_i}{v_h} = -\frac{V_c}{2v_h} + \sqrt{\left(\frac{V_c}{2v_h}\right)^2 + 1} = 0.87353$$

$$\therefore v_i = 9.6613 \text{ (m/s)}$$

$$\therefore \lambda_i = 0.045927$$

$$\therefore \lambda = \lambda_i + \lambda_c = 0.060188$$

$$\therefore C_T = \frac{2\rho A (V_c + v_i) v_i}{\rho A \Omega^2 R^2} = 2\lambda \lambda_i = 5.52851 \times 10^{-3} \quad * C_T \text{ remains unchanged as } T = W$$

$$\therefore \theta_0 = 0.239 \text{ (rad)} = 13.70^\circ$$

$$\therefore C_p = 3.8138 \times 10^{-4}$$

$$\therefore P = 398408 \text{ (W)}$$

Answers:

1. 871.8 kW
2. Derivation, $w = 2v_i$
3. (a) 1,426 kW
(b) 1,729 kW
4. Derivation
5. (a) $\theta = 13.05^\circ$, 354 kW
(b) $\theta = 13.74^\circ$, 401.6 kW